Tools Used and Tools Planned for this Uganda Fake News Detection Project

A. Tools Used at the Current Stage

1. Data Collection Tools

Manual Web Collection

- AfricaCheck
- PesaCheck
- Mainstream news websites (Daily Monitor, New Vision, BBC Africa, VOA, CNN, Reuters)

Reason:

- Ensured high-quality, verified labels
- Avoided noisy or weakly labeled data
- Appropriate for a first version of a domain-specific African dataset

2. Development Environment

- Python 3
- UV (Python environment & dependency management)
- Jupyter Notebook

Reason:

- Reproducible experiments
- Clean dependency isolation
- Interactive analysis for EDA and modeling

3. Data Handling & Analysis

- Pandas
- NumPy

Purpose:

- Dataset loading and cleaning
- Handling missing values
- Label encoding
- Data inspection and summarization

4. Exploratory Data Analysis (EDA)

- Matplotlib
- Seaborn

Used for:

- Class distribution plots (Fake vs True)
- Source distribution
- Language distribution
- Text length analysis

Outcome:

- Confirmed class balance
- Identified language mix (English / Swahili)
- Verified dataset consistency

5. Text Preprocessing

- NLTK
    - ✓ Tokenization
    - ✓ Stopword removal
- Regex (re)
    - ✓ Text normalization
- Scikit-learn preprocessing utilities

Purpose:

- Noise reduction
- Standardization before feature extraction

6. Feature Extraction

- TF-IDF Vectorizer (scikit-learn)

Why TF-IDF:

- Strong baseline for text classification
- It is efficient and interpretable
- It is widely used in academic research

7. Machine Learning Model (Baseline)

- Logistic Regression (scikit-learn)

Why Logistic Regression:

- Interpretable
- Strong baseline for binary classification
- Easy to benchmark against advanced models

8. Model Evaluation

- Scikit-learn metrics

    - ✓ Accuracy
    - ✓ Precision
    - ✓ Recall
    - ✓ F1-Score
    - ✓ Classification report

Current Performance:

- Accuracy ≈ 93%
- Balanced precision and recall

B. Tools Planned for Next Phases

1. Additional Classical ML Models

- Scikit-learn

    - ✓ Support Vector Machine (SVM)
    - ✓ Multinomial Naive Bayes
    - ✓ Random Forest

Purpose:

- Comparative evaluation
- Identify best-performing classical model

2. Deep Learning Frameworks

- PyTorch or TensorFlow / Keras

Models:

- LSTM / Bi-LSTM
- CNN for text classification

Purpose:

- Capture deeper semantic patterns
- Compare with classical ML

3. Transformer-Based NLP Models

- Hugging Face Transformers
- Pretrained Models

    - ✓ BERT
    - ✓ DistilBERT
    - ✓ XLM-RoBERTa (for multilingual text)

Purpose:

- Handle English–Swahili mixed data
- Improve performance on African news content

4. Model Explainability

- SHAP or LIME

Purpose:

- It explain predictions
- Improve trust and transparency
- Useful for academic and policy discussions

5. System Prototype

- Flask or FastAPI
- React Js
- Appwrite
- PostgreSQL

Purpose:

- Demonstrate real-world application
- User inputs news text → model outputs classification

6. Version Control & Reproducibility

- Git
- GitHub

Purpose:

- Track experiments
- Ensure reproducibility
- Facilitate academic review

In Conclusion

So far, I've used Python with Pandas and scikit-learn for data analysis, TF-IDF for feature extraction, and Logistic Regression as a baseline model. For the next phase, I plan to compare additional ML models and then fine-tune transformer models like BERT or XLM-RoBERTa, followed by building a simple prototype system.